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**Department of
Digital Innovation and
Technology**

INSTRUCTIONS TO CANDIDATES

1. Answer **ALL** questions in Section A
2. Answer any **FOUR (4)** out of five (5) questions in Section B.
3. Write **ALL** answers on the answer book (and MC answer sheet, if any) provided.
4. This question paper has **SEVENTEEN (17)** pages.

Programme Code:	DS125109 / DS145109
Module Title:	Enterprise Architecture & System Development
Module Code:	DIT5307

**End-of-Semester Examination
Semester Two 2024/25**

Date : 19 May 2025
Time : 2:00 p.m. – 5:00 p.m.
Time Allowed : 3 hours

Section B (80 marks) – Answer ANY FOUR (4) questions

INSTRUCTIONS ON FILLING IN MISSING SOURCE CODE

For some parts in this exam paper, you are required to fill in missing source code. For example, if it is required that the square of a variable x be returned in a certain method and the incomplete code is "return (iv)____;" then you should write down the following answer in the answer book.

(iv) $x * x$

If you see two or more blanks with the same Roman numeral e.g.:

```
System.(iii)____.println("Hello Student.");
```

```
System.(iii)____.println("Welcome to this game! ;) ");
```

It means that they have the same content, so you just need to write your answer once for them. In this example you should write

(iii) out

Question: B1

- (a) Describe what is meant by *enterprise application software*.
What are the characteristics of an enterprise application?
(4 marks)
- (b) Java Platform, Enterprise Edition (Java EE) uses a multi-tiered application model for enterprise applications.
- (i) Draw a diagram to show the different tiers of this application model, clearly marking which tier(s) correspond(s) to the Java EE server, the database server and the client machine respectively, as well as showing the type of component in each tier. (6 marks)
 - (ii) Describe briefly the components corresponding to the Java EE server. (4 marks)
- (c) Refer to the following code snippets from three source files login.html, greeting.jsp and GreetingServlet.java of a certain Java Web application.

login.html

```
...  
<form action="greet" method="post">  
    ...  
    <label>User name:</label>  
    <input type="text" name="username" required>  
    ...  
</form>
```

greeting.jsp

```
...  
<p>Hello ${username}, Good ${time},.</p>  
...
```

(... continued on next page)

GreetingServlet.java

```
...
@WebServlet("/greet")
public class GreetingServlet extends HttpServlet {
    ...
    @Override
    protected void doPost((i)_____request,
                           HttpServletResponse response)
        throws ServletException, IOException {
        String timeText = "morning";
        String inputName = (ii)_____;
        if (...) // now is between 12:00 and 17:59
            timeText = "afternoon";
        else if (...) // now is between 18:00 and 23:59
            timeText = "evening";

        (iii)_____((iv)_____, timeText);
        (iii)_____((v)_____, inputName);

        getServletContext()
            . (vi)_____ ("/greeting.jsp")
            . forward (request, response);
    }
    ...
}
```

Fill in the missing source code.

(6 marks)

Question: B2

(a) Suggest the best type of enterprise bean for each of the following scenarios. Explain your suggestion.

- (i) An online e-commerce platform which sells hundreds of different types of items provides a shopping cart feature which lets a user virtually 'put' items in or 'remove' items from the cart while browsing different products.
- (ii) When a user of a global air ticket booking website is viewing air ticket, the website shows the total number of users who are also viewing the same flight.
- (iii) A currency web portal shows the current price of a foreign currencies after the user enters the exchange code.

(9 marks)

(b) To implement the shopping cart mentioned above in part (a)(ii), the development team has decided to code an enterprise bean with remote access. Also, as a good practice, the bean will implement a business interface. Refer to the source code given on the next page and answer the following questions:

- (i) Write a Java annotation which should replace the comment `/* annotation missing here */`. (1 mark)
- (ii) Write the interface declaration for `ShoppingCart`, including all necessary annotation(s). You are NOT required to write package and import statements. (6 marks)
- (iii) One of the team members suggests that `getAll()` method just needs to return `SelectedItem` but do nothing else. Do you agree? Why? (4 marks)

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Question: B2 (b) (cont.)

```
...
(i)/* annotation missing here */
public class CartEJB implements ShoppingCart, Serializable {
    ...
    private List<SelectedItem> selectedItems;

    @PostConstruct
    public void doInit() {
        ...
        selectedItems = new ArrayList<>();
    }

    @Override
    public void put(String itemID, int Qty) {
        ...
    }

    @Override
    public void remove(String itemID, int Qty) {
        ...
    }

    @Override
    public List<SelectedItem> getAll() {
        List<SelectedItem> results = new ArrayList<>();
        results.addAll(selectedItems);

        return results;
    }

    @Remove()
    @Override
    public void clearAll() {
        selectedItems = null;
    }
}
```

Question: B3

- (a) Compare and contrast a message-driven bean with a stateless session bean. (10 marks)
- (b) What is the advantage of using message-driven beans to receive messages in a Java EE server? (4 marks)
- (c) Source code snippets of a message-driven bean which receives messages sent to a destination with JNDI name `jms/THEiQueue` in a point-to-point style is given below.

```
...
(i) _____(activationConfig = {
    @ActivationConfigProperty(
        propertyName = (ii)_____,
        propertyValue = "jms/THEiQueue"
    ),
    @ActivationConfigProperty((iii)_____)
})
public class THEiMessageBean (iv)_____ {
    public THEiMessageBean() {
    }
    @Override
    (v)_____ (Message message) {
        // code for processing message goes here
    }
}
```

Fill in the missing source code.

(6 marks)

Question: B4

- (a) Refer to some simplified JSP source code implementing a shopping cart of a legacy online e-commerce application as follows.

```
<body>
  <%@ page import="java.util.List, ..." %>
  ...
  <%
    List selectedItems=(List)request.getAttribute("items");
  %>
  <table>
    <tr>
      <th>Item Description</th>
      <th>Unit Price</th>
      <th>Quantity</th>
    </tr>
    <%
      for (SelectedItem item : selectedItems) {
    %>
    <tr>
      <td><%= item.getItemDescription()%></td>
      <td><%= item.getUnitPrice()%></td>
      <td><%= item.getQuantity()%></td>
    </tr>
    <%
      } // end for
    %>
  </table>
  ...
</body>
```

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Question: B4 (a) (cont.)

Since it is now considered bad practice to code scriptlets (code within `<% %>`), you are required to rewrite the code shown in **bold** above with EL and JSTL so that all `<% %>` scriptlets and `<%= %>` expressions are removed. You can safely assume that Book class is a legitimate JavaBeans component and you are NOT required to include the `taglib` directive that specifies the use of the JSTL core library in your answer. (10 marks)

- (b) To better comply with the Model-View-Controller (MVC) pattern, the online e-commerce platform decides to use JSF technology. Rewrite the code in part (a) with Facelets. Assume that the backing bean shown below has been properly coded already. (10 marks)

```
...
@Named
@SessionScoped
public class ShoppingCart implements Serializable {
    ...
    public List<SelectedItem> getCartItems () {
        // Return a List of all items in the cart
        ...
    }
    ...
}
```

Question: B5

An entity class Address is used to represent a typical address in Hong Kong. Field id is used as the primary key, whose value is provided automatically by the database. To be a valid address, there must be something in the street and territory (HK, KL or NT) fields and territory must be of length 2. An address is associated with another entity class Person such that a person entity can have one or more addresses. Consider the following source code snippets:

Address.java

```
...
import java.io.Serializable;
import javax.persistence.*;
import javax.(i)_____.*;
(ii)_____
(iii)_____ {
    (iv)_____
    private String street;
    private Long id;
    private String flat;
    private int floor;
    private String block;
    private String estate;
    @NotBlank
    (v)_____
    private String district;
    @(vi)_____ (vii)_____, (viii)_____)
    private String territory;
    private boolean isPrimary;
    (ix)_____
    private Person person;
    private boolean active;

    // Constructor, setter and getter methods, etc
    ...
}
```

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Question: B5 (cont.)

Person.java

```
...
import javax.persistence.*;
...
..... Person ..... {
    ...
    (x) _____
    private Collection<Address> addresses;
    ...
}
```

AddressManager.java

```
...
import javax.ejb.Stateless;
import javax.persistence.*;
@Stateless
public class AddressManager {
    (xi) _____
    private (xii) _____ em;

    public void save(Address address) {
        em.persist(address);
    }

    public void delete(Address address) {
        em.(xiii) _____(address);
    }
    // Other methods go here ...
}
```

- (a) Fill in the missing source code. (16 marks)
- (b) Use a comparison table to show any four different between Java EE and Spring Framework. (4 marks)

--- End of Paper ---